

Ex. 1: Simulation of FDMA in MATLAB

Code:

```
clear; close all; clc; %% Parameters
Fs = 600e3;
t = 0:1/Fs:5e-3;
fm = 5e3;
fc = 20e3;
m = sin(2*pi*fm*t);
c = cos(2*pi*fc*t);
s = m .* c;
figure;
subplot(4,1,1);
plot(t(1:1000), m(1:1000), 'LineWidth',1.3);
title('Message Signal (5 kHz)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,2);
plot(t(1:1000), c(1:1000), 'LineWidth',1.3);
title('Carrier Signal (20 kHz)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(4,1,3);
plot(t(1:1000), s(1:1000), 'LineWidth',1.3);
title('Modulated Signal (Message x Carrier)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
N = length(s);
S = abs(fft(s)/N);
f = (-N/2:N/2-1)*(Fs/N);
subplot(4,1,4);
plot(f, fftshift(S), 'LineWidth',1.3);
title('Spectrum of Modulated Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
xlim([0 40e3]);
[~, idx] = max(fftshift(S));
peak_freq = f(idx);
fprintf("\nSpectrum Peak Detected at = %.1f kHz\n", peak_freq/1000);
```

